

Cauchy-characteristic matching for the Z4c evolution system: method

z4c-CMM verified research loop¹

¹generated from the mission-2 knowledge ledger (*knowledge-database/paper_z4c-CCM/*);
every equation and number below is traceable to a verifier output under *results/numerical/*
(Dated: June 13, 2026)

I. INTRODUCTION

Cauchy-characteristic matching (CCM) couples a strongly hyperbolic interior (Cauchy) evolution to a characteristic evolution reaching future null infinity, so that the worldtube supplying characteristic boundary data simultaneously receives the incoming radiative content computed on the characteristic grid; the interface becomes transparent rather than approximately absorbing [1, 2]. The present paper formulates and verifies CCM for the Z4c formulation of the Einstein equations [3–5]: the constraint-damped conformal decomposition in production use across finite-difference numerical-relativity codes, here the AthenaK implementation [6]. The presentation follows the organization of Ref. [1]: the Cauchy system and its boundary sector (Sec. II), the characteristic system (Sec. III), the matching algorithm (Sec. IV), and numerical verification (Sec. V). Every equation is one of: (a) a machine-verified result of this program (dark blue; verification labels [Vn] and the ledger nodes N1–N16); (b) standard literature content, cited at the equation; (c) an explicitly open obligation (labels O-Nx-y). The derivation chain is summarized as a directed acyclic graph in Fig. 1.

II. THE Z4C CAUCHY SYSTEM AND ITS BOUNDARY SECTOR

Black equations are standard literature content, cited at the equation; **dark blue** equations are the machine-verified new Z4c-CCM content (verification labels [Vn], Table II; ledger nodes in section titles).

A. Boundary frame and Z4c variables

$$ds^2 = -\alpha^2 dt^2 + \gamma_{ij} (dx^i + \beta^i dt)(dx^j + \beta^j dt), \quad n^\mu = \alpha^{-1} (1, -\beta^i), \quad (1)$$

$$s_i = \frac{\partial_i r}{\sqrt{\gamma^{rr}}}, \quad s^i = \frac{\gamma^{ir}}{\sqrt{\gamma^{rr}}}, \quad m^A = \frac{1}{\sqrt{2}} (e_2 + i e_3)^A, \quad (2)$$

$$\mathring{l}^\mu = \frac{1}{\sqrt{2}} (n^\mu + s^\mu), \quad \mathring{k}^\mu = \frac{1}{\sqrt{2}} (n^\mu - s^\mu) \quad [5]. \quad (3)$$

Z4c variables [5]:

$$\chi = (\det \gamma)^{-1/3}, \quad \tilde{\gamma}_{ij} = \chi \gamma_{ij}, \quad K = \hat{K} + 2\Theta, \quad \tilde{A}_{ij} = \chi \left(K_{ij} - \frac{1}{3} \gamma_{ij} K \right), \quad (4)$$

$$\partial_t \chi = \frac{2}{3} \chi [\alpha (\hat{K} + 2\Theta) - D_i \beta^i], \quad (5)$$

$$\partial_t \tilde{\gamma}_{ij} = -2\alpha \tilde{A}_{ij} + \beta^k \tilde{\gamma}_{ij,k} + 2\tilde{\gamma}_{k(i} \beta_{j)}^k - \frac{2}{3} \tilde{\gamma}_{ij} \beta^k_{,k} \quad [5]. \quad (6)$$

B. The Z4c evolution system

The Z4c formulation [3, 4] evolves the conformal variables (4) together with $\hat{K} = K - 2\Theta$, the conformal trace-free curvature $\tilde{A}_{ij}$, the constraint pair (Θ, Z_i) — with Z_i carried by $\tilde{\Gamma}^i = \tilde{\gamma}^{jk} \tilde{\Gamma}_{jk}^i + 2\tilde{\gamma}^{ij} Z_j$ — and the gauge (α, β^i) . In vacuum, with D_i the γ -covariant derivative and $[\cdot]^\text{TF}$ the γ -trace-free part, the evolution equations are [3]

$$\partial_t \hat{K} = \beta^i \partial_i \hat{K} - D^i D_i \alpha + \alpha \left[\tilde{A}_{ij} \tilde{A}^{ij} + \frac{1}{3} (\hat{K} + 2\Theta)^2 \right] + \alpha \kappa_1 (1 - \kappa_2) \Theta, \quad (7)$$

$$\partial_t \tilde{A}_{ij} = \beta^k \partial_k \tilde{A}_{ij} + 2 \tilde{A}_{k(i} \partial_{j)} \beta^k - \frac{2}{3} \tilde{A}_{ij} \partial_k \beta^k + \chi [\alpha R_{ij} - D_i D_j \alpha]^{\text{TF}} + \alpha \left[(\hat{K} + 2\Theta) \tilde{A}_{ij} - 2 \tilde{A}_{ik} \tilde{A}^k_j \right], \quad (8)$$

$$\partial_t \Theta = \beta^i \partial_i \Theta + \frac{\alpha}{2} \left[\mathcal{R} + \frac{2}{3} (\hat{K} + 2\Theta)^2 - \tilde{A}_{ij} \tilde{A}^{ij} \right] - \alpha \kappa_1 (2 + \kappa_2) \Theta, \quad (9)$$

$$\begin{aligned} \partial_t \tilde{\Gamma}^i &= \beta^k \partial_k \tilde{\Gamma}^i - \tilde{\Gamma}^k \partial_k \beta^i + \frac{2}{3} \tilde{\Gamma}^i \partial_k \beta^k + \tilde{\gamma}^{jk} \partial_j \partial_k \beta^i + \frac{1}{3} \tilde{\gamma}^{ij} \partial_j \partial_k \beta^k \\ &\quad - 2 \tilde{A}^{ij} \partial_j \alpha + 2\alpha \left[\tilde{\Gamma}^i_{jk} \tilde{A}^{jk} - \frac{3}{2} \tilde{A}^{ij} \partial_j \ln \chi - \frac{1}{3} \tilde{\gamma}^{ij} \partial_j (2\hat{K} + \Theta) - \kappa_1 \tilde{\gamma}^{ij} Z_j \right], \end{aligned} \quad (10)$$

with $\mathcal{R}$ the Ricci scalar of γ_{ij} and (κ_1, κ_2) the constraint-damping parameters (production: $\kappa_1 = 0.02$, $\kappa_2 = 0$). The gauge is 1+log slicing and the gamma-driver shift [3, 6],

$$\partial_t \alpha = \beta^i \partial_i \alpha - \mu_L \alpha^2 \hat{K}, \quad \partial_t \beta^i = \mu_S \tilde{\Gamma}^i - \eta \beta^i + \beta^j \partial_j \beta^i, \quad (11)$$

(AthenaK realization: $\mu_L = 2/\alpha$, $\mu_S = 1$, $\eta = 2$ [6]). The constraints are

$$\mathcal{H} = \mathcal{R} + \frac{2}{3} K^2 - \tilde{A}_{ij} \tilde{A}^{ij}, \quad \mathcal{M}_i = D^j (K_{ij} - \gamma_{ij} K), \quad \Theta = 0, \quad Z_i = 0, \quad (12)$$

and at the principal level the constraint subsystem propagates as waves, $\square \Theta \simeq 0 \simeq \square Z_i$ [5], the fact underlying the boundary rows of Sec. IIF and the stability analysis of node N13.

C. The ten incoming boundary modes

At a timelike outer boundary with frame (1)–(3), the linearized Z4c system carries ten incoming characteristic fields (taxonomy verified at nodes N4/N5: transverse-traceless = physical; h_{ss} = gauge; h_{sA} = constraint-vector; tangential trace = constraint-scalar): The constraint and gauge rows are those of Ref. [5] adapted to the Z4c variable set; the

Table I. Incoming boundary modes of Z4c-CCM and their data sources (ledger node N6). Per-speed outgoing characteristic vectors $\hat{X}^a = (\hat{n}^a + \sqrt{v} \hat{s}^a)/\sqrt{2}$ at sector speed v [5]; production order $L \geq 1$.

#	sector	incoming field	speed	boundary condition (datum)
1–2	physical	$U_{AB}^- = \text{TT}_2[E + \epsilon(s)B]_{AB}$	1	$U_{AB}^- \hat{=} -(\psi'_0 \tilde{m}_A \tilde{m}_B + \text{c.c.})$ [CCM: $\psi'_0 = A^2 \psi_0^{\text{CCE}}$]
3	constraint	Θ	1	$(r^2 \hat{l} \cdot \partial)^{L+1} \Theta \hat{=} 0$
4	constraint	Z_s	1	$(r^2 \hat{l} \cdot \partial)^{L+1} Z_s \hat{=} 0$
5–6	constraint	Z_A	1	$(r^2 \hat{l} \cdot \partial)^{L+1} Z_A \hat{=} 0$
7	gauge	α	$\sqrt{\mu_L}$	$(r^2 \hat{m} \cdot \partial)^{L+1} \alpha \hat{=} h_\alpha$
8	gauge	β_s	$\sqrt{\nu_s}$	$(r^2 \hat{k}_{\nu_s} \cdot \partial)^{L+1} \beta_s \hat{=} h_s$
9–10	gauge	β_A	$\sqrt{\mu_S}$	$(r^2 \hat{j} \cdot \partial)^{L+1} \beta_A \hat{=} h_A$

physical rows 1–2 carry the CCM datum and constitute the new content (Secs. IID–IIE, IIF). With $\psi'_0 \rightarrow 0$ the set reduces exactly to the absorbing constraint-preserving scheme of Ref. [5] [V6]. The frozen-coefficient LF/GKS analysis of the constraint rows (node N13) shows that every matched-falloff advection realization of rows 3–6 is stable and absorbs the (Θ, Z_s) pair at $1 - O((\sigma/2\omega r)^2)$, that falloff-mismatched sink variants self-generate $O(10\%)$ pair reflection, and that the production default (the stock matched realization) is the admissible optimum.

D. Tetrad correspondence and Type-II boost (node N2)

Outgoing null label u with $u|_{\partial\Omega} = t$: $\partial_\mu u = (1, u_r, 0, 0)$,

$$g^{tt} + 2g^{tr} u_r + g^{rr} u_r^2 = 0 \quad \implies \quad u_r = \frac{-g^{tr} - \sqrt{(g^{tr})^2 - g^{tt} g^{rr}}}{g^{rr}}, \quad u_r|_{\text{flat}} = -1. \quad (13)$$

Characteristic outgoing null vector [1, 2]:

$$\hat{l}^\mu = -e^{2\hat{\beta}} g^{\mu\nu} \partial_\nu u. \quad (14)$$

For arbitrary $(\alpha, \beta^i, \gamma_{ij})$,

$$\hat{l}^\mu = \frac{\sqrt{2}}{2} (\alpha - \gamma_{ij} \beta^i \beta^j) e^{-2\hat{\beta}} \hat{l}^\mu \quad [\text{V1: exact at 12 rational points; GPU residual } 9.3 \times 10^{-16}], \quad (15)$$

$$A = (\alpha - \gamma_{ij}\beta^i s^j) e^{-2\hat{\beta}}, \quad \psi'_0 = A^2 \psi_0, \quad (16)$$

i.e. the Choice-2 boost of Ref. [1] holds verbatim on the Z4c frame (3): no evolution-system variable enters (15). ψ_0 on the characteristic grid [1, 2]:

$$\psi_0 = \left(\frac{r\partial_r\beta - 1}{4Kr} \right) \left[(1+K)\partial_r J - \frac{J^2\partial_r\bar{J}}{1+K} \right] + \frac{J(1+K^2)\partial_r J \partial_r\bar{J}}{8K^3} + \frac{1}{8K} \left[\frac{J^2\partial_r^2\bar{J}}{1+K} - (1+K)\partial_r^2 J \right] - \frac{J\bar{J}^2(\partial_r J)^2 + J^3(\partial_r\bar{J})^2}{16K^3}. \quad (17)$$

E. Physical-mode dictionary (node N3, exact)

Electric/magnetic Weyl decomposition w.r.t. n^μ (vacuum):

$$E_{\mu\nu} = C_{\mu\alpha\nu\beta} n^\alpha n^\beta, \quad B_{\mu\nu} = \frac{1}{2} \epsilon_{\mu\alpha}{}^{\gamma\delta} C_{\gamma\delta\nu\beta} n^\alpha n^\beta, \quad \epsilon_{\mu\nu\rho\sigma} = \sqrt{-g} [\mu\nu\rho\sigma]. \quad (18)$$

EXACT identity (pure Weyl algebra, generic 10-component tensor; no perturbation, no field equations) [V2a]:

$$(E_{\mu\nu} + \epsilon_\mu{}^{\kappa\lambda} s_\kappa B_{\lambda\nu}) m^\mu m^\nu = -\psi_0, \quad (E_{\mu\nu} - \epsilon_\mu{}^{\kappa\lambda} s_\kappa B_{\lambda\nu}) m^\mu m^\nu = -\bar{\psi}_4, \quad (19)$$

with $\psi_0 = -C_{abcd} l^a m^b l^c m^d$ on the frame (3); dyad completeness ($U_{AB} = U(m, m) \bar{m}_A \bar{m}_B + \text{c.c.}$ for tangential trace-free U) gives the tensor form of the incoming Weyl object

$$U_{AB}^- \equiv \text{TT}_2[E + \epsilon(s)B]_{AB} = -(\psi_0 \bar{m}_A \bar{m}_B + \bar{\psi}_0 m_A m_B). \quad (20)$$

Realization from the Z4c state (Gauss–Codazzi, on-shell vacuum; γ_{ij}, K_{ij} from (4)) [V2c]:

$$E_{ij} = R_{ij}[\gamma] + K K_{ij} - K_{ik} K^k{}_j, \quad B_{ij} = \epsilon_i{}^{kl} D_k K_{lj}|_{\text{sym}}, \quad \epsilon_{ikl} = \sqrt{\det \gamma} [ikl]. \quad (21)$$

Nonperturbative verification (Kerr–Schild Schwarzschild, autodiff, both routes; Table II V2c): $|E^{3+1} - E^{4D}|$ rel. 7.7×10^{-17} , $|B^{3+1} - B^{4D}|$ rel. 3.9×10^{-17} , identity residual 8.5×10^{-16} . Linearized on-shell flat limit (superseded defining form, now a corollary [V2b]): for TT $h_{AB}(t, x_s)$, $\psi_0 = \frac{1}{4}(\partial_t + \partial_s)^2 h_{mm}$ and $U_{AB}^- \rightarrow -\frac{1}{4}(\partial_t + \partial_s)^2 \gamma_{AB}^{\text{TF}}$, i.e. $(\partial_t + \partial_s)^2 \gamma_{AB}^{\text{TF}} = 4(\psi_0 \bar{m}_A \bar{m}_B + \text{c.c.})$.

F. The Z4c-CCM boundary conditions (node N6)

Ten incoming modes at the worldtube; $X_v^\mu = (n^\mu + \sqrt{v} s^\mu)/\sqrt{2}$ at sector speed v ; $L \geq 1$:

$$\text{physical (2):} \quad U_{AB}^- \hat{=} -(\psi'_0 \bar{m}_A \bar{m}_B + \bar{\psi}'_0 m_A m_B), \quad \psi'_0 = A^2 \psi_0^{\text{CCE}} \quad (\text{exact, (20)–(21)}), \quad (22)$$

$$\text{constraint (4):} \quad (r^2 \mathring{l} \cdot \partial)^{L+1} \Theta \hat{=} 0, \quad (r^2 \mathring{l} \cdot \partial)^{L+1} Z_s \hat{=} 0, \quad (r^2 \mathring{l} \cdot \partial)^{L+1} Z_A \hat{=} 0, \quad (23)$$

$$\text{gauge (4):} \quad (r^2 \mathring{m} \cdot \partial)^{L+1} \alpha \hat{=} 0, \quad (r^2 \mathring{k}_{\nu_s} \cdot \partial)^{L+1} \beta_s \hat{=} 0, \quad (r^2 \mathring{j} \cdot \partial)^{L+1} \beta_A \hat{=} 0. \quad (24)$$

Björhus form of (22) (the correction drives the incoming Weyl object to its CCM value, the Z4c analog of Ref. [1]):

$$U_{AB}^- - U_{AB}^-|_{\text{BC}} \rightarrow 0, \quad U_{AB}^-|_{\text{BC}} = -(\psi'_0 \bar{m}_A \bar{m}_B + \bar{\psi}'_0 m_A m_B); \quad \psi_0^{\text{CCE}} \rightarrow 0 \implies \text{CPBC scheme of Ref. [5]} \quad [\text{V6}]. \quad (25)$$

Equations (23)–(24) are the order- L conditions of Ref. [5]; (24) replaces the Sommerfeld gauge placeholder of Ref. [1].

G. Sector orthogonality (nodes N4, N5)

For arbitrary TT data $h_{AB}(t, x_s)$ [V4]:

$$\mathcal{H} = \partial_i \partial_j h_{ij} - \partial^2 h^k{}_k = 0, \quad \mathcal{M}_i = \partial_j \dot{h}_{ij} - \partial_i \dot{h}^j{}_j = 0, \quad \gamma^{ij} \partial_t h_{ij} = 0, \quad \tilde{\Gamma}^i = -\partial_j \tilde{h}^{ij} = 0; \quad (26)$$

negative controls: $h_{x_s y} = G(t, x_s) \Rightarrow \mathcal{M}_y = -\frac{1}{2} \partial_{x_s} \dot{G} \neq 0$; $h_{yy} = h_{zz} = G \Rightarrow \mathcal{H} = -2 \partial_{x_s}^2 G \neq 0$.

Hence the datum (22) sources neither (23) nor (24) at linear order: the three sectors of (22)–(24) commute.

On the boundary wave model of Ref. [5], $[\partial_t^2 - 2\dot{\beta} \partial_t \partial_x - (1 - \dot{\beta}^2) \partial_x^2] u = 0$, plane-wave reflection [V5]:

$$R_{\text{Sommerfeld}} = -\frac{\dot{\beta}(1 - \dot{\beta})}{(1 + \dot{\beta})(2 - \dot{\beta})} \neq 0, \quad R_{(\partial_t + c_+ \partial_x)} = 0 \quad (\text{identically}). \quad (27)$$

H. Frozen-coefficient boundary stability (node N7)

Laplace–Fourier ansatz $u = \tilde{u} e^{st + \lambda x}$, $\Re s > 0$ [V7]:

$$\lambda_+ = \frac{s}{1 + \dot{\beta}} \text{ (admissible)}, \quad \lambda_- = -\frac{s}{1 - \dot{\beta}}, \quad D_L(s) = \left(s + (1 + \dot{\beta}) \lambda_+ \right)^{L+1} = (2s)^{L+1}, \quad \frac{|D_L(s)|}{|s|^{L+1}} = 2^{L+1}. \quad (28)$$

D_L contains no datum symbol: the injection (22) modifies only the inhomogeneity, so the boundary-stability results of Ref. [5] (constraint sector: all L ; gauge+metric sector: spherical reduction, asymptotically harmonic shift) transfer verbatim with $\|u\| \leq C \|\psi'_0\text{-datum}\|$.

III. THE CHARACTERISTIC SYSTEM

A. Bondi–Sachs framework

The exterior of the worldtube $r = r_{\text{wt}}$ is evolved on outgoing null slices in Bondi–Sachs form [2, 7, 8],

$$ds^2 = -\left(e^{2\beta} \frac{V}{r} - r^2 h_{AB} U^A U^B \right) du^2 - 2e^{2\beta} du dr - 2r^2 h_{AB} U^B du dx^A + r^2 h_{AB} dx^A dx^B, \quad (29)$$

with the spin-weighted scalars built from the complex dyad q^A ($q^A q_A = 0$, $q^A \bar{q}_A = 2$): the radiative variable $J = \frac{1}{2} q^A q^B h_{AB}$, the lapse-like β , the shift-like $U = q_A U^A$ with its radial momentum $Q = r^2 e^{-2\beta} q^A h_{AB} \partial_r U^B$, the mass-like $W = (V - r)/r^2$, and the time derivative $H = \partial_u J$ at fixed compactified radius [2]. The radial coordinate is compactified as

$$y = 1 - \frac{2r_{\text{wt}}}{r}, \quad y \in [-1, 1], \quad \frac{dy}{dr} = \frac{(1 - y)^2}{2r_{\text{wt}}}, \quad (30)$$

so the worldtube sits at $y = -1$ and future null infinity at $y = 1$. On each null slice the Einstein equations order into the radial hypersurface hierarchy $\beta \rightarrow Q \rightarrow U \rightarrow W \rightarrow H$ (each equation an ODE in r sourced by the previous members and J), followed by the evolution equation $\partial_u J = H$ [1, 2, 7]; we do not reproduce the nonlinear system here and work instead with its linearization, which the program verifies term by term.

B. Linear $\ell = 2$ scalarization

At linear order about Minkowski, the axisymmetric $(\ell, m) = (2, 0)$ sector reduces to one real scalar per field [machine-derived and verified, ledger node N14]: with $s \equiv \sin \theta$, $c \equiv \cos \theta$, $P \equiv 3c^2 - 1$,

$$J = J_t s^2, \quad Q = Q_t s c, \quad U = U_t s c, \quad W = W_t P, \quad \beta = \beta_t P, \quad H = H_t s^2, \quad (31)$$

and the $\ell = 2$ spin-weighted derivatives close on the same shapes,

$$\bar{\partial} \beta = -6 \beta_t s c, \quad \bar{\partial} \bar{\partial} \beta = -6 \beta_t P, \quad \bar{\partial}^2 \beta = 6 \beta_t s^2, \quad \bar{\partial} Q = -Q_t s^2, \quad \bar{\partial} U = -U_t s^2. \quad (32)$$

The spin-weighted spherical-harmonic modal coefficients used for code interoperation follow from

$$s^2 = 4 \sqrt{\frac{2\pi}{15}} {}_2Y_{20}, \quad s c = -\sqrt{\frac{8\pi}{15}} {}_1Y_{20}, \quad P = \sqrt{\frac{16\pi}{5}} Y_{20}. \quad (33)$$

In the Bondi gauge ($\beta = 0$) the verified linear hierarchy is

$$\partial_r (r^2 Q_t) = -4 r^2 \partial_r J_t, \quad \partial_r U_t = \frac{Q_t}{r^2},$$

$$\partial_r(r^2 W_t) = 2J_t + 2rU_t + \frac{r^2}{2}\partial_r U_t, \quad \partial_r(rH_t) = -\frac{U_t}{2} + \frac{r}{4}\partial_r^2 J_t - \frac{3J_t}{2r}, \quad (34)$$

where the H row is the plain-family pinned combination; the joint exact-rational refit on the enlarged (gauge-shifted) solution family fixes the equivalent covariant form (35) below [derivation artifact `derive_beta_hierarchy.jl`].

C. The β -corrected hierarchy for live (anchored-gauge) data

Worldtube data constructed from a live Cauchy state arrive in the worldtube-anchored gauge of Sec. IV, which carries $\beta \neq 0$ (r -independent at this order) and non-decaying gauge tails. The hierarchy sources required by such data were machine-derived as exact-rational fits on the anchored and plain solution families jointly:

$$\begin{aligned} \partial_r(r^2 Q) &= -r^2 \partial_r(\bar{\partial} J) - 4r \bar{\partial} \beta, & \partial_r U &= Q/r^2, \\ \partial_r(r^2 W) &= \frac{1}{2} \bar{\partial} \bar{\partial} J + 2r \bar{\partial} U + \frac{r^2}{2} \partial_r(\bar{\partial} U) - \frac{4}{3} \bar{\partial} \bar{\partial} \beta, & \partial_r(rH) &= \frac{1}{2} r \partial_r^2 J + \partial_r J + \frac{\bar{\partial}^2 \beta}{r} - \frac{\bar{\partial} Q}{2r} - \bar{\partial} U. \end{aligned} \quad (35)$$

Scalarized with (31)–(32) and recast on the compactified coordinate (30), the system becomes four ODEs with coefficients regular on the whole interval $y \in [-1, 1]$ in *both* gauges (the $y = 1$ rows are automatic regularity conditions):

$$\begin{aligned} 2Q_t + (1-y) Q'_t &= -4(1-y) J'_t + 24\beta_t, & U'_t &= \frac{Q_t}{2r_{\text{wt}}}, \\ 2W_t + (1-y) W'_t &= \frac{(1-y)J_t}{r_{\text{wt}}} + 2U_t + \frac{(1-y)U'_t}{2} + \frac{4\beta_t(1-y)}{r_{\text{wt}}}, & & \\ H_t + (1-y) H'_t &= \frac{(1-y)}{4r_{\text{wt}}} \left[(1-y)^2 J''_t - 2(1-y) J'_t \right] + \frac{(1-y)^2}{2r_{\text{wt}}} J'_t + \frac{3\beta_t(1-y)}{r_{\text{wt}}} + \frac{Q_t(1-y)}{4r_{\text{wt}}} + U_t, \end{aligned} \quad (36)$$

with $' = d/dy$. The *ringed* variables rJ_t , r^2U_t , $\dots$ diverge at $y = 1$ on anchored-gauge data (the $k = 0, 1$ tail monomials); the unringed form (36) is the representation in which the anchored solution is finite everywhere [measured failure of the ringed scheme on anchored data: ψ_0 off by $O(10^{3-4})$; pinned as a regression test].

D. Spectral solver, ψ_0 extraction, and pulse data

The radial direction is discretized on Chebyshev–Gauss–Lobatto nodes with the standard differentiation matrix; the linear solves of (36) use LU factorizations built once, each operator’s boundary row imposing the worldtube value. Time advance is RK4 on $\partial_u J_t = H_t$ with a penalty (SAT) imposition of the worldtube Dirichlet value,

$$H_t|_{y=-1} = \tau (J_t^{\text{wt}} - J_t|_{y=-1}), \quad \tau = \frac{\sigma N^2}{2r_{\text{wt}}}, \quad \sigma = 1. \quad (37)$$

The incoming Weyl datum is extracted from J_t alone,

$$\psi_0^t(r) = -\frac{\partial_r J_t}{2r} - \frac{\partial_r^2 J_t}{4}, \quad \psi_0 = \psi_0^t s^2, \quad (38)$$

an expression *gauge-invariant on the anchored family*: the anchored tails $a(u) + b(u)/r$ satisfy $\psi_0^t[a + b/r] = 0$ identically, and the machine check $\psi_0[J_{\text{anchored}}] = \psi_0[J_{\text{Bondi}}]$ holds as an exact rational identity [N14]. Evaluation at an interior radius $r_* \geq r_{\text{wt}}$ (required by the matching geometry of Sec. IV) uses barycentric interpolation at $y_* = 1 - 2r_{\text{wt}}/r_*$ together with

$$\partial_r J_t = \frac{(1-y)^2}{2r_{\text{wt}}} J'_t, \quad \partial_r^2 J_t = \frac{(1-y)^2}{4r_{\text{wt}}^2} \left[(1-y)^2 J''_t - 2(1-y) J'_t \right]; \quad (39)$$

the interior point is *better* conditioned than the worldtube node (measured: evolved-probe relative error 8.8×10^{-5} vs the $\sim 6 \times 10^{-2}$ worldtube floor at the same resolution and step). For the pulse-injection test of Sec. V the initial null slice carries the compactly supported $\ell = 2$ free datum specified in the test design of Ref. [1] (polynomial envelope times Gaussian in y , amplitude Z , support $[y_{\min}, y_{\max}]$, center y_c , width τ_y), converted to the t -scalar convention by

$$J_t(y)|_{u_0} = \frac{1}{4} \sqrt{\frac{15}{2\pi}} \mathcal{J}(y), \quad \mathcal{J}(y) = \frac{4Z (y_{\max} - y)(y - y_{\min})}{(y_{\max} - y_{\min})^2} e^{-(y-y_c)^2/\tau_y^2} \text{ on } [y_{\min}, y_{\max}], \quad (40)$$

and zero outside the support.

E. Exact Teukolsky reference in Bondi variables

The solver verifier is the exact linear Teukolsky solution (Sec. VB, Eqs. (51)–(54)) transformed to the Bondi gauge; the machine-derived closed forms are

$$\begin{aligned} J &= \frac{3}{4} s^2 \left[\frac{F^{(4)}}{r} + \frac{F^{(2)}}{r^3} \right], & \beta &= 0, & H &= \partial_u J, \\ Q &= sc \left[\frac{3F^{(4)}}{r} - \frac{9F^{(3)}}{r^2} - \frac{9F^{(2)}}{r^3} \right], & U &= sc \left[-\frac{3F^{(4)}}{2r^2} + \frac{3F^{(3)}}{r^3} + \frac{9F^{(2)}}{4r^4} \right], \\ W &= P \left[-\frac{3F^{(4)}}{2r^2} - \frac{3F^{(3)}}{2r^3} - \frac{3F^{(2)}}{4r^4} \right], \end{aligned} \quad (41)$$

all arguments retarded, $F^{(n)} = F^{(n)}(u)$ with $u = t - r$. The hierarchy (34) closes on (41) as a rational identity; the incoming datum is

$$\psi_0 = -\frac{9}{8} s^2 \frac{F^{(2)}(u)}{r^5} \quad (\text{exactly; no subleading retarded terms}), \quad (42)$$

consistent with (56), and the outgoing waveform is the finite-radius expression (59) with the convention map stated there. An independent production characteristic code run on worldtube data built from (41) reproduces $\Psi_4 = -\sqrt{6\pi/5} F^{(6)}$ at future null infinity to 6×10^{-6} relative and (42) to 0.1% (Sec. V).

F. Solver verification summary

(i) Exact-slice sweeps of (36) are machine-exact ($\sim 10^{-22}$) on both the Bondi and anchored families. (ii) The SAT evolution reproduces the worldtube ψ_0 at the documented double-precision conditioning floor and the interior probe (39) at 8.8×10^{-5} relative (7.1×10^{-6} at halved step; truncation-dominated), identically for plain and anchored boundary data. (iii) The native C++ production port is gated against the verification-harness numbers (standalone test; relative agreement $\sim 10^{-7}$). (iv) The pulse problem (40) is radially converged at $n = 65$ collocation points to 2.1×10^{-7} of peak (resolution ladder), with time-stepping error 4×10^{-12} . (v) The constraint-row boundary analysis and the live two-way coupling are treated in Secs. II and IV–V respectively.

IV. THE MATCHING ALGORITHM

The matching loop closes two data paths each Cauchy step: the worldtube path (Cauchy state $\rightarrow$ characteristic boundary data, Secs. IV A–IV C) and the datum path (characteristic $\psi_0 \rightarrow$ the physical boundary rows of Sec. IIF, Secs. IV D–IV E). The full derivation chain is summarized in Fig. 1.

A. Worldtube data map (node N1)

$$g_{00} = -\alpha^2 + \gamma_{ij} \beta^i \beta^j, \quad g_{0i} = \gamma_{ij} \beta^j, \quad g_{ij} = \gamma_{ij}; \quad g^{00} = -\alpha^{-2}, \quad g^{0i} = \frac{\beta^i}{\alpha^2}, \quad g^{ij} = \gamma^{ij} - \frac{\beta^i \beta^j}{\alpha^2}. \quad (43)$$

ADM kinematics requires $\partial_t \gamma_{ij} = -2\alpha K_{ij} + \beta^k \partial_k \gamma_{ij} + \gamma_{kj} \partial_i \beta^k + \gamma_{ik} \partial_j \beta^k$. From $\gamma_{ij} = \chi^{-1} \tilde{\gamma}_{ij}$,

$$\partial_t \gamma_{ij} = \chi^{-1} \partial_t \tilde{\gamma}_{ij} - \chi^{-2} \tilde{\gamma}_{ij} \partial_t \chi, \quad (44)$$

and inserting (5)–(6) with $K_{ij} = \chi^{-1} (\tilde{A}_{ij} + \frac{1}{3} \tilde{\gamma}_{ij} K)$, $\beta^k \partial_k \gamma_{ij} = \chi^{-1} \beta^k \partial_k \tilde{\gamma}_{ij} - \gamma_{ij} \beta^k \partial_k \ln \chi$,

$$[\partial_t \gamma_{ij}]_{\text{Z4c}} - [\partial_t \gamma_{ij}]_{\text{ADM}} = \gamma_{ij} \left[\frac{2}{3} D_k \beta^k - \frac{2}{3} \partial_k \beta^k + \beta^k \partial_k \ln \chi \right]. \quad (45)$$

$D_i \beta^i = \partial_k \beta^k - \frac{3}{2} \beta^k \partial_k \ln \chi$

[V3: closure exact; naive $D = \partial_k \beta^k$ leaves residual $+1 \cdot \gamma_{ij} \beta^k \partial_k \ln \chi$]. (46)

B. Live sampling and the $\ell = 2$ data contract

The Cauchy ADM state is interpolated to three geodesic spheres at r_{wt} and $r_{\text{wt}} \pm \delta r$ each step. With orthonormal frame components $h_{\hat{a}\hat{b}} = (\gamma - \delta)_{\hat{a}\hat{b}}$ and $\partial_t h_{\hat{a}\hat{b}} = -2K_{\hat{a}\hat{b}}$ (linear order; $\alpha = 1 + O(h)$, the correction entering beyond linear), the $(\ell, m) = (2, 0)$ contract scalars are the shape projections

$$h_{\text{TT}} = \left\langle \frac{1}{2}(h_{\hat{\theta}\hat{\theta}} - h_{\hat{\phi}\hat{\phi}}) \right\rangle_{s^2}, \quad h_{rr} = \langle h_{\hat{r}\hat{r}} \rangle_P, \quad h_{\text{tr}} = \langle h_{\hat{\theta}\hat{\theta}} + h_{\hat{\phi}\hat{\phi}} \rangle_P, \quad h_{r\theta} = \langle r h_{\hat{r}\hat{\theta}} \rangle_{sc}, \quad (47)$$

with $\langle f \rangle_X = \int f X d\Omega / \int X^2 d\Omega$ for $X \in \{s^2, P, sc\}$, the time derivatives obtained from the same projections of $-2K$, and $\partial_r h_{\text{tr}}$ by the centered difference across the sphere triplet. At $\ell = 2$ the angular derivatives close analytically on the same scalars,

$$\langle \partial_\theta h_{\text{tr}} \rangle_{sc} = -6 h_{\text{tr}}, \quad \langle \partial_\theta \partial_r h_{\text{tr}} \rangle_{sc} = -6 \partial_r h_{\text{tr}}, \quad \langle (\partial_\theta + \cot \theta) h_{r\theta} \rangle_P = h_{r\theta}, \quad (48)$$

so the contract is seven numbers per step: $(h_{\text{TT}}, \partial_t h_{\text{TT}}, h_{rr}, h_{\text{tr}}, \partial_r h_{\text{tr}}, h_{r\theta}, \partial_t h_{r\theta})$.

C. The anchored worldtube map (six rows + β)

The Cauchy coordinates near the worldtube do not satisfy the Bondi gauge conditions; the radial gauge generator is fixed by areal-radius consistency only up to functions of (u, θ) . Anchoring the generator to vanish *at the worldtube*, with the angular generator corrected so that the $g_{r\theta}$ gauge condition holds consistently ($\xi_a = \xi_{\text{old}} + \partial_\theta(\zeta_u^{\text{wt}})/r - \text{const}$), makes the map from the contract (47)–(48) to the characteristic boundary scalars *local*:

$$\begin{aligned} J_t^{\text{wt}} &= h_{\text{TT}}, & H_t^{\text{wt}} &= \partial_t h_{\text{TT}}, & U_t^{\text{wt}} &= \frac{6 h_{\text{tr}}}{4 r_{\text{wt}}}, \\ Q_t^{\text{wt}} &= \frac{6 h_{\text{tr}}}{4} + \frac{6 r_{\text{wt}} \partial_r h_{\text{tr}}}{4} - \frac{3 h_{r\theta}}{r_{\text{wt}}} + \partial_t h_{r\theta}, & W_t^{\text{wt}} &= \frac{3 h_{rr}}{4 r_{\text{wt}}} - \frac{\partial_r h_{\text{tr}}}{4}, & \beta_t^{\text{wt}} &= \frac{h_{rr}}{4} - \frac{h_{\text{tr}}}{8} - \frac{r_{\text{wt}} \partial_r h_{\text{tr}}}{8} + \frac{h_{r\theta}}{4 r_{\text{wt}}}, \end{aligned} \quad (49)$$

verified as an exact identity against the anchored-gauge solution series (BigFloat-192 residual $< 10^{-35}$; derivation artifact `derive_wtmap_v2.jl`). The anchored gauge necessarily carries $\beta_t \neq 0$ and non-decaying tails $a(u) + b(u)/r$ in J, U, W ; these are exactly the structures supported by the β -corrected, unringed system (36), and they are ψ_0 -silent by (38). Independent anchoring of the radial and angular generators violates the $g_{r\theta}$ condition and is excluded [pinned: the uncorrected map fails the β channel with a characteristic $(4 - n)$ -ratio pattern].

D. Retarded matching geometry

The characteristic time is the retarded cone label, $u = t - r$ on the flat background: the cone u meets the worldtube worldline at Cauchy time $t = u + r_{\text{wt}}$ and the Cauchy outer boundary r_B at $t = u + r_B$. Hence

$$\text{tube sample at } t \longrightarrow \text{cone } u = t - r_{\text{wt}}, \quad \text{boundary datum at } t \longleftarrow \psi_0^t(r_B) \Big|_{u=t-r_B}, \quad (50)$$

a causality lag of $r_B - r_{\text{wt}}$ in u that is always available in lockstep evolution. The solver records the interior probe (39) at r_B each substep and serves the lagged value by linear interpolation in the recorded history, with the analytic initial-data value as the pre-cone fallback (the run starts on the quiescent cone $u_0 = -r_{\text{wt}}$). Boundary data are held fixed across the substeps between Cauchy samples — an $O(\Delta t)$ coupling approximation. Measured against the published reference data of Ref. [1] (Sec. V), this approximation, together with the tube-inside-datum geometry, produces $O(30\%)$ deviations of the live boundary ψ_0 from the clean characteristic solution in the pulse-arrival window; linear-in-time boundary-data interpolation is the corresponding open refinement [O-M5-1, class (c)].

E. Injection path, cadence, and initialization

The served datum enters the Cauchy evolution through the Type-II boost and the physical rows of Sec. IIF: $\psi'_0 = A^2 \psi_0$ with $A = (\alpha - \gamma_{ij} \beta^i s^j) e^{-2\beta}$ (16), injected as (58). Per Cauchy RK stage at the boundary: (1) sample the contract (47) (collective over ranks; off-rank sphere points are excluded by ownership, the shape norms being global); (2) map by (49); (3) advance the characteristic state to $u = t - r_{\text{wt}}$ (root rank); (4) serve the lagged probe (50)

and broadcast; (5) apply the boundary rows. The characteristic initial slice is the analytic datum (quiescent for the matched tests; (40) for the pulse-injection test); initialization order and the $\hat{\beta}$ circularity at $t = 0$ remain documented obligations [O-N6-3]. Setting the served datum to zero reduces the entire loop to the absorbing scheme of Ref. [5] exactly [V6].

F. Derivation chain

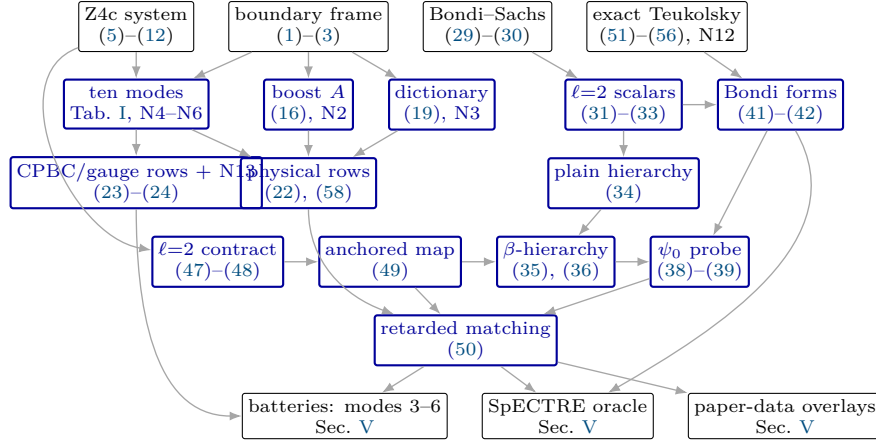


Figure 1. Derivation chain of the Z4c-CCM formulation. Black nodes: foundations (literature content and the exact reference chain); blue nodes: machine-verified new content (ledger nodes named in the section titles); bottom row: the verification instruments of Sec. V.

V. NUMERICAL VERIFICATION

A. Numerical evidence

All runs: JAX float64, $4 \times$ NVIDIA A100-80GB (CUDA_VISIBLE_DEVICES=0,1,2,3), wall clock ≤ 23 s each (budget 600s); outputs under `results/numerical/`.

Table II. Symbolic and sampling verification of (46)–(28). “Exact” = sympy identity on arbitrary functions or exact rationals (zero float error). Files: `n2_boost_check.txt`, `n3_exact_check.txt`, `n3_dictionary_check.txt`, `n1_varmap_check.txt`, `n4_cpbc_compat_check.txt`, `n5_gauge_check.txt`, `n6_composite_check.txt`, `n7_lf_sketch_check.txt`.

verified statement	exact part	GPU sweep (4 devices)
V1 boost identity (15), factor $\sqrt{2}/2$	12/12 rational points	9.34×10^{-16} over 4,194,304 samples
V2a EXACT identity (19)–(20), $(\sigma, \kappa) = (1, -1)$, $\bar{\psi}_4$ partner	generic Weyl, all 10 dof	—
V2b linearized on-shell limit (corollary)	arbitrary F_1, F_2	0.0 over 4,194,304 modes (prior form)
V2c Gauss–Codazzi realization (21), Schwarzschild	Codazzi sign +1 derived	E : 7.7×10^{-17} , B : 3.9×10^{-17} , identity 8.5
V3 maps (43), closure (46)	full-symbol identities	round trip 4.10×10^{-16} , gg^{-1} 8.88×10^{-16}
V4 orthogonality (26) + controls	identities and non-vanishing controls	TT residual 0.0 vs control 12.3
V5 reflection (27)	exact R algebra	$ R_{\text{Som}} \geq 0.023$ on $\hat{\beta} \in [0.05, 0.6]$; $R = 0$ ex
V6 10-mode count, $\psi_0 \rightarrow 0$ reduction, wire-through (exact targets)	structural	reduction 0.0; chain residual 1.47×10^{-16}
V7 Kreiss condition (28), $L = 0..3$	$D_L = (2s)^{L+1}$ exact	ratio deviation 3.2×10^{-14} (4,194,304 sam

B. Dynamical test at the parameters of Ref. [1] (nodes N10–N11)

Full 3+1 evolution: AthenaK Z4c [6] (vendored, pinned; RK4, CFL 0.25, sixth-order stencils at `nghost` = 4, $\kappa_1 = 0.02$), with (22) realized at the Cauchy outer boundary as described below. This subsection is self-contained: every symbol used by the test is defined here or in Secs. II A–II F.

Table III. Dynamical model evolution (6th-order interior stencils with a GKS-stable mixed closure — 4th-order Fornberg rows at three edge nodes; fully one-sided 6th-order closures are RK4-unstable (error ledger) — RK4, CFL 0.2, shift $\hat{\beta} = 0.3$, outgoing Gaussian pulse onto the boundary; one configuration per GPU; `n8_model1d_test.txt`). Reflection is measured in the L_2 (energy) norm — trapezoid sums of Gaussian pulses converge exponentially, so the deviation column is limited by machine arithmetic, not by measurement or truncation. Analytic L_2 relation: $\|q_{\text{in}}\|_{L_2}/\|q_{\text{out}}\|_{L_2} = \frac{\hat{\beta}}{2+\hat{\beta}} \sqrt{\frac{1+\hat{\beta}}{1-\hat{\beta}}}$. Each row sits at its float64 floor: rows 1–3 at the per-sample roundoff floor (deviation saturates between $N=8193$ and $N=16385$: 1.9 vs 2.1×10^{-14}); the injection row at the linear roundoff-accumulation optimum ($\sim \text{steps} \times \varepsilon$; ladder $N=16385/32769/65537/131073 \Rightarrow 5.4/2.4/5.9/30 \times 10^{-11}$, non-monotone and insensitive to CFL 0.2 vs 0.05).

boundary condition	measured	exact	deviation
Sommerfeld, L_2 ratio ($N=16385$)	0.177752646227	0.177752646227	2.1×10^{-14} (float64 floor)
Sommerfeld, L_2 ratio ($N=8193$)	—	—	1.9×10^{-14} (saturated)
$(\partial_t + c_+ \partial_x)$ [P1/CCM op., $N=16385$]	$R_{L_2} = 1.9 \times 10^{-14}$	0	machine precision
<u>CCM injection (25), rel. L_2 ($N=32769$)</u>	<u>2.4×10^{-11}</u>	<u>0</u>	<u>roundoff-accumulation optimum</u>

a. Exact solution injected. Even-parity $(\ell, m) = (2, 0)$ Teukolsky wave [9] with the Gaussian profile of Ref. [1], written with physicists' Hermite polynomials H_n and the origin-regular retarded/advanced combinations:

$$F(u) = X e^{-(u-r_c)^2/\tau^2}, \quad F^{(n)}(u) = X \left(-\frac{1}{\tau}\right)^n H_n\left(\frac{u-r_c}{\tau}\right) e^{-(u-r_c)^2/\tau^2}, \quad (51)$$

$$Q^{(n)}(t, r) = F^{(n)}(t-r) - (-1)^n F^{(n)}(t+r), \quad R^{(n)}(t, r) = F^{(n)}(t-r) + (-1)^n F^{(n)}(t+r), \quad \partial_t Q^{(n)} = R^{(n+1)}. \quad (52)$$

Radial functions (regular at $r=0$ by (52); calligraphic to keep A for the boost (16)):

$$\begin{aligned} \mathcal{A} &= 3 \left[\frac{Q^{(2)}}{r^3} + \frac{3Q^{(1)}}{r^4} + \frac{3Q^{(0)}}{r^5} \right], & \mathcal{B} &= - \left[\frac{Q^{(3)}}{r^2} + \frac{3Q^{(2)}}{r^3} + \frac{6Q^{(1)}}{r^4} + \frac{6Q^{(0)}}{r^5} \right], \\ \mathcal{C} &= \frac{1}{4} \left[\frac{Q^{(4)}}{r} + \frac{2Q^{(3)}}{r^2} + \frac{9Q^{(2)}}{r^3} + \frac{21Q^{(1)}}{r^4} + \frac{21Q^{(0)}}{r^5} \right], \end{aligned} \quad (53)$$

orthonormal-frame metric perturbation ($\dot{h}_{ij}$ from $Q^{(n)} \rightarrow R^{(n+1)}$ termwise):

$$h_{\hat{r}\hat{r}} = \mathcal{A} \frac{1 + 3 \cos 2\theta}{2}, \quad h_{\hat{r}\hat{\theta}} = -3\mathcal{B} \sin \theta \cos \theta, \quad h_{\hat{\theta}\hat{\theta}} = 3\mathcal{C} \sin^2 \theta - \mathcal{A}, \quad h_{\hat{\phi}\hat{\phi}} = -3\mathcal{C} \sin^2 \theta + \mathcal{A} (3 \sin^2 \theta - 1), \quad (54)$$

$$\gamma_{ij}(0, x) = \delta_{ij} + h_{ij}(0, x), \quad K_{ij}(0, x) = -\frac{1}{2} \dot{h}_{ij}(0, x) \quad (\text{linear order; Ref. [1] solves XCTS}). \quad (55)$$

Weyl scalars of the solution at linear order (the self-datum and the extraction reference):

$$r^5 \psi_0(t, r, \theta) = -\sqrt{\frac{27\pi}{10}} F^{(2)}(t-r) {}_2Y_{20}(\theta), \quad {}_2Y_{20} = \frac{1}{4} \sqrt{\frac{15}{2\pi}} \sin^2 \theta, \quad r \psi_4|_{(2,0)} \rightarrow -\sqrt{\frac{6\pi}{5}} F^{(6)}(t-r). \quad (56)$$

b. Discrete realization of (22). At every Cauchy-boundary point, after the CPBC/gauge rows (23)–(24) (Sommerfeld task of Ref. [6]), the two physical rows are driven through $\tilde{A}_{ij}$:

$$s^a = \frac{x^a/r}{\sqrt{\gamma_{bc} x^b x^c/r}}, \quad e_\phi \propto \hat{z} \times s, \quad e_\theta = s \times e_\phi, \quad \bar{m} = \frac{1}{\sqrt{2}} (e_\theta - i e_\phi), \quad (57)$$

$$\partial_t \tilde{A}_{ab}|_{\partial\Omega} = -\frac{2\chi}{\alpha} \left(\psi'_0 \bar{m}_a \bar{m}_b + \bar{\psi}'_0 m_a m_b \right), \quad \psi'_0 = A^2 \psi_0^{\text{datum}}, \quad A = (\alpha - \gamma_{ij} \beta^i s^j) e^{-2\hat{\beta}} \quad (58)$$

with ψ_0^{datum} the right-hand member of (56) evaluated at the boundary point ($\hat{\beta} = 0$ on the flat background); $\psi_0^{\text{datum}} \rightarrow 0$ reduces (58) to the unmodified scheme exactly (V6; machine-zero reduction in Table III methodology, AthenaK battery: Minkowski preserved to 2.6×10^{-33}).

c. Exact finite-radius reference (node N12). The extraction reference is the *analytic* waveform at the extraction radius itself (no auxiliary reference run): in the AthenaK extraction convention ($= -1 \times$ the scri member of (56) continued to finite r),

$$r \psi_4^{(2,0)}(t, r) = \sqrt{\frac{6\pi}{5}} \left[F^{(6)} + \frac{2F^{(5)}}{r} + \frac{3F^{(4)}}{r^2} + \frac{3F^{(3)}}{r^3} + \frac{3F^{(2)}}{2r^4} \right] (t-r) - \sqrt{\frac{6\pi}{5}} \frac{3F^{(2)}}{2r^4} (t+r), \quad (59)$$

verified by seven symbolic gates (tracelessness, coefficient-exact linearized vacuum, $\sin^2 \theta$ factorization, peeling, scri limit, independent 30-digit finite-difference cross-check at 10^{-40}) and — independently — by the `SpECTRE CharacteristicExtract` cross oracle below.

d. Protocol and result. Parameters of Ref. [1]: $r_c = 20$, $\tau = 2$, $X = 10^{-5}$, Cauchy boundary (worldtube analog) at 41, $t \in [0, 80]$; extraction of $r\psi_4^{(2,0)}$ at $r = 36$; resolution ladder $h = 0.5, 0.331, 0.277$ ($4 \times A100$, every run < 600 s). Error against (59) at each rung's own output times:

$$E(h) = \frac{\|r\psi_4^{(2,0)}[h] - r\psi_4^{(2,0)}[\text{exact}]\|_2}{\|r\psi_4^{(2,0)}[\text{exact}]\|_2} = 4.99 \times 10^{-2}, \quad 5.36 \times 10^{-3}, \quad 2.02 \times 10^{-3}, \quad (60)$$

with measured convergence order 5.39 (first pair) and 5.52 (second pair), finest-rung amplitude $1 \pm 2 \times 10^{-4}$ and time shift 0.000 (Fig. 2). Two production defects were found *only* by this exact-reference program and are corrected in all data shown: a frame-orientation error in the initial data ($\hat{e}_\theta$ flipped; linearized vacuum violated at 5% of $|R_{ij}|$, invisible to constraint monitors and to run-vs-reference protocols since it sat on both sides) and a one-step offset in the waveform time labels. The self-datum CCM–Sommerfeld difference remains peeling-suppressed ($\psi_0 \sim r^{-5}$, relative imprint $\sim 10^{-9}$ of peak at this worldtube), as in Ref. [1] Sec. V.A.

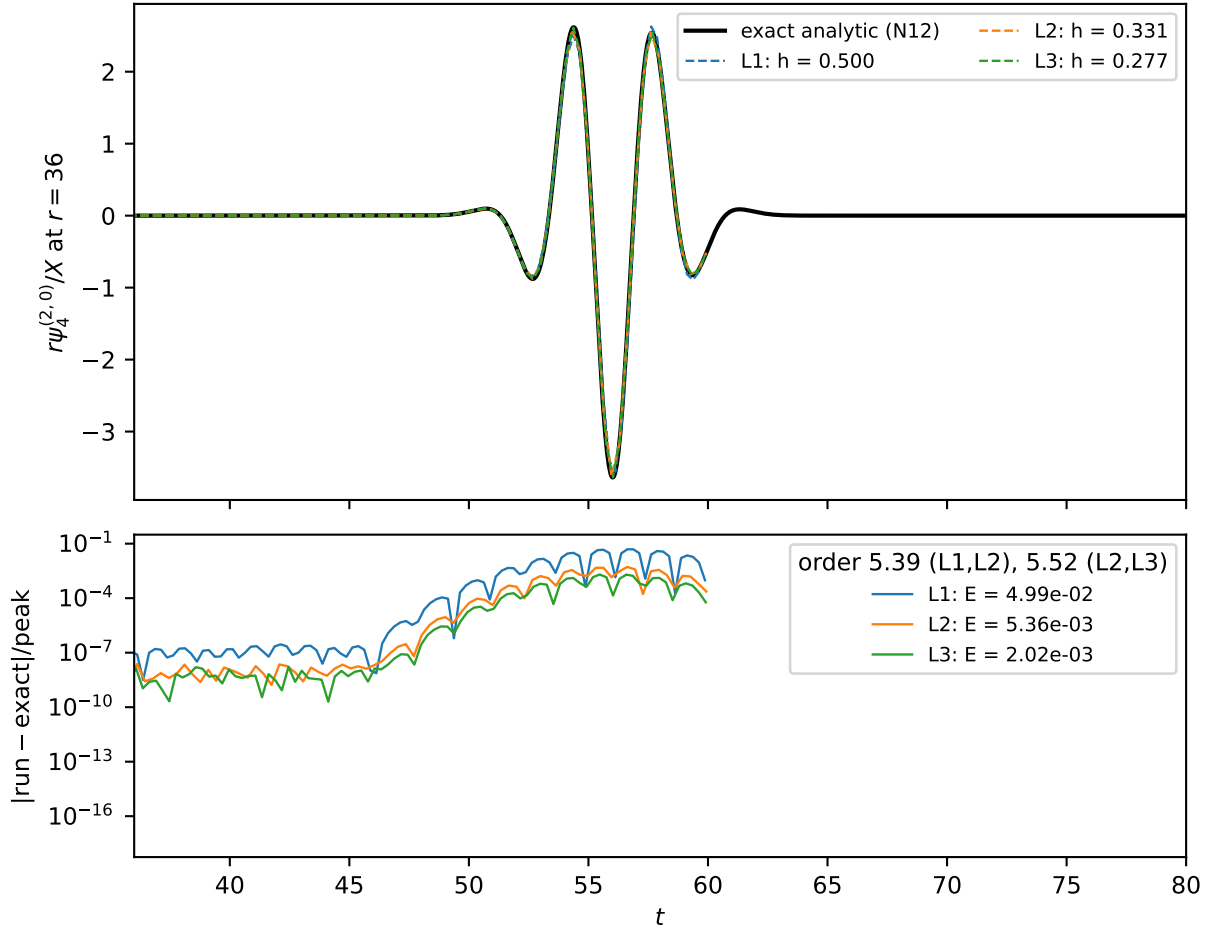


Figure 2. Exact-reference Teukolsky test of Sec. VB ($r_c=20$, $\tau=2$, boundary 41, extraction $r=36$, $X=10^{-5}$). Top: $(\ell, m)=(2, 0)$ mode of $r\psi_4$ — the exact finite-radius waveform (59) (solid) and the three-rung sixth-order ladder. Bottom: pointwise errors against the exact reference; measured order 5.39/5.52, finest-rung relative L_2 error 2.02×10^{-3} (60). Generator: `scripts/plot_paper_figures_v2.py`; data: `results/numerical/athenak_teuk_ana/gpu6/` (ledger node N12).

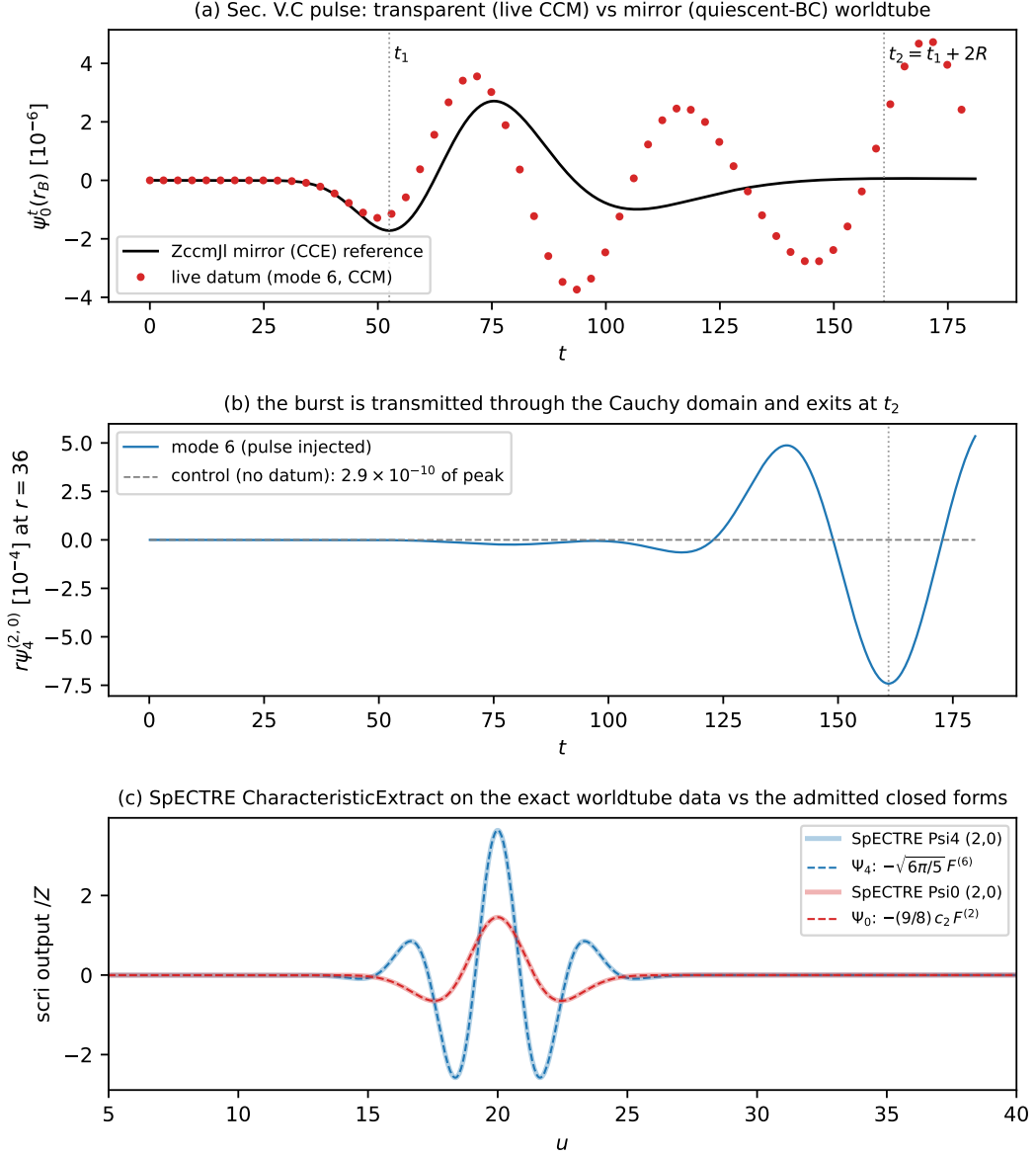


Figure 3. Live-CCM verification (Secs. III C–III D, IV). (a) The characteristic-pulse fidelity test: live ψ_0 datum at the Cauchy boundary vs. the quiescent-worldtube (mirror) reference — early-window agreement 7×10^{-8} of peak; the later $O(1)$ separation is the matching physics. (b) The transmitted burst at the extraction radius exits at $t_2 = t_1 + 2R$; the no-datum control is consistent with zero. (c) SpECTRE CharacteristicExtract on the exact worldtube data (41) vs. the closed forms. Generator: `scripts/plot_paper_figures_v2.py`; data: `results/numerical/n14_native/`, `n14_pulse_ref.csv`, `spectre_oracle/`.

VI. CONCLUSIONS

-
- [1] S. Ma *et al.*, Fully relativistic three-dimensional Cauchy-characteristic matching for physical degrees of freedom (2023), local source: [ref-paper/arxiv-2308.10361/](#), [arXiv:2308.10361 \[gr-qc\]](#).
 - [2] J. Moxon, M. A. Scheel, and S. A. Teukolsky, Improved Cauchy-characteristic evolution system for high-precision numerical relativity waveforms (2020), local source: [ref-paper/arxiv-2007.01339/](#), [arXiv:2007.01339 \[gr-qc\]](#).
 - [3] S. Bernuzzi and D. Hilditch, Phys. Rev. D **81**, 084003 (2010), [0912.2920](#).
 - [4] D. Hilditch, S. Bernuzzi, M. Thierfelder, Z. Cao, W. Tichy, and B. Bruegmann, Phys. Rev. D **88**, 084057 (2013), [1212.2901](#).

- [5] M. Ruiz, D. Hilditch, and S. Bernuzzi, Constraint preserving boundary conditions for the Z4c formulation of general relativity (2010), local source: `ref-paper/``arxiv-1010.0523v2/`, [arXiv:1010.0523 \[gr-qc\]](#).
- [6] H. Zhu *et al.*, Performance-portable numerical relativity with AthenaK (2024), local source: `ref-paper/``arxiv-2409.10383/`; vendored fork pinned in `athenak/PROVENANCE.md`, [arXiv:2409.10383 \[gr-qc\]](#).
- [7] H. Bondi, M. G. J. van der Burg, and A. W. K. Metzner, Proc. Roy. Soc. Lond. A **269**, 21 (1962).
- [8] R. K. Sachs, Proc. Roy. Soc. Lond. A **270**, 103 (1962).
- [9] S. A. Teukolsky, Phys. Rev. D **26**, 745 (1982).